

Beyond Collusion Resistance: Leveraging Social Information for Plural Funding and Voting

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Abstract

In its 2018 introduction[2] and subsequent work[8], Quadratic Funding (QF) emerged as a uniquely optimal design for the democratic provision of public goods under the assumption of atomized participants with perfect rationality. In this paper, we aim to move past this rational and atomized portrayal of human beings and explore pluralistic QF mechanisms that recognize the influence of social connections among participants and incentivize cooperation across social differences. With this goal in mind, we define *collusion resistance* as a criterion against disproportional power accumulation in Quadratic Funding models that can arise from pre-existing participant relationships and argue that collusion resistant QF and plural QF are two sides of the same coin. Next, we evaluate various iterations of Quadratic Funding, testing their collusion resistance and other social and technical issues. Finally, we propose several new mechanisms, including Connection-Oriented Cluster Match, which satisfies our definition of collusion resistance. Our findings show great potential for making Quadratic Funding more pluralistic. They might also guide principles and practices of computational design that bridge the epistemological divides between classical economics and social reality.

1 Introduction

Since its introduction in 2018[2], Quadratic Funding (QF) along with its older cousin Quadratic Voting (QV) [12], have shown promise as technologies capable of revolutionizing governance and providing alternatives to standard capitalist political economies.

QF is a mechanism for allocating funds to public goods. As input, QF takes contributions (in some unit of account) to a slate of projects. Say individuals 1 through n contribute c_1 through c_n to a project: then QF gives that project

$$\left(\sum_{i=1}^n \sqrt{c_i} \right)^2 \quad (1)$$

in funding, with externally provided matching funds making up the (always positive) difference between this sum and the simple linear sum of contributions. Figure 1 shows a geometric interpretation of the formula, which we will return to throughout this paper. Figure 1 also illustrates how the total amount of funding awarded by QF is made up partly of the contributions of individual agents and partly of an extra *matching subsidy* awarded by the system from an external fund.

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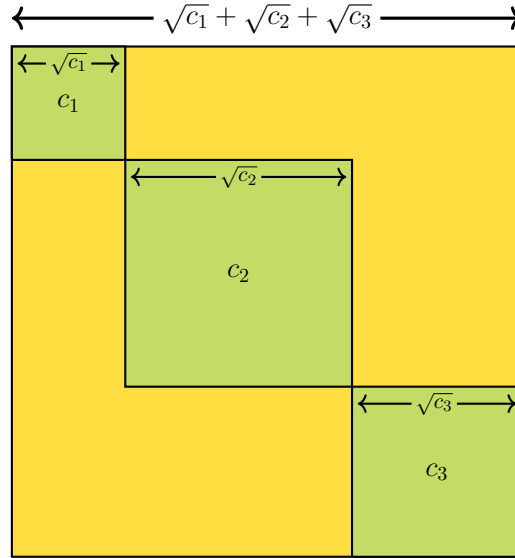


Figure 1: A visual representation of QF with three contributions. This diagram (and others like it throughout this paper) represents a contribution c_i as a green square with area c_i . The yellow areas represent the *matching subsidy* added by QF. The area of the large square is the total funding amount awarded by QF.

[2] shows that under standard mechanism design assumptions (e.g., quasi-linear utility, selfish utility-maximizing agents) and complete information, the equilibrium of QF maximizes utilitarian social welfare; [8] shows it does so uniquely among the class of mechanisms that have a “aggregative” form. In other words, QF funds projects until society as a whole would not be better off increasing funding even a tiny amount. In practice, private, primarily blockchain-based initiatives like Gitcoin [9] and clr.fund [4] already implement QF to allocate capital to Open Source software communities, to the tune of more than \$70 million in grants so far. Public institutions (e.g., the executive [7] and legislative [16] government of Colorado and the government of Taiwan [19]) also implement QV at slightly smaller scales.

Moreover, one can consider QF part of a more significant academic thrust to study and implement *plural* social technologies [20] under what has been called the Plurality paradigm. Plurality does not see people as atomized, rational agents but instead as part a rich web of social connections. It seeks to recognize those socio-cultural associations as the basis for various social technologies like financial systems, governance schemes or property relations to help people cooperate across their differences. Accordingly, one of the fundamental ideas behind pluralism is to build technologies that can coordinate communities while respecting the diversity and social complexity within those communities. QF is one such technology, partly because of its potential to prioritize cooperation across differences, also known as “consilience”. One way to see this potential is to notice that a project with many different supporters collectively contributing some amount will receive much more funding than a project with only one supporter who contributes that same amount on their own. Figure 2 gives visual intuition for this fact. Under the Plurality paradigm, QF is appealing because it rewards projects liked by distinct individuals while also leaving space for people to express the strength of their preferences. Notably, small groups could still out-vote larger groups –

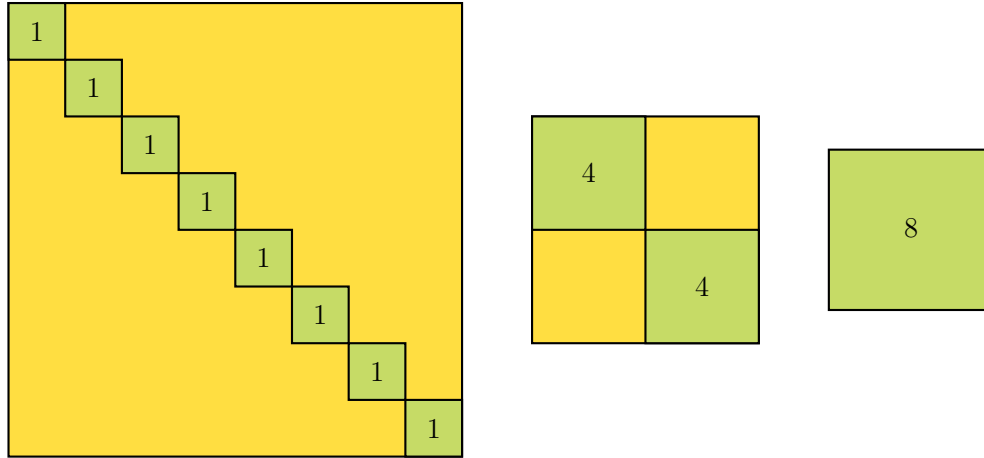


Figure 2: Three different QF diagrams drawn to scale with each other. In each case, the sum of contributions is \$8. Note that eight agents who contribute \$1 each yield more funding than two agents contributing \$4 each, which in turn yields more funding than one agent contributing \$8.

they would have to contribute more capital to do so, i.e., express more passionate preferences.

Moving from models to the real world always brings new challenges, and one practical challenge facing QF is its vulnerability to collusion. For example, what if instead of contributing \$8 to a project that they benefited from, someone were able to create eight Sybil accounts and give a unit from each of them? In this case, the funding level would look like the leftmost diagram in figure 2 when really it should look like the rightmost diagram in that same figure. For QF to work in practice, surely there needs to be some way to sniff out that type of foul play. Now suppose a company did something similar by mandating that each of its eight employees give a unit to a project that benefited the company. This case smells like collusion, but simple Sybil resistance wouldn't be enough to stop it, unlike in the first example.

On the other hand, there are clear differences between eight Sybils contributing capital and eight coordinated employees. One could also imagine those eight employees contributing independently simply out of shared interest in the project (rather than due to top-down command). Here, any "collusion" would be in good faith. So how do we make sense of all these differences and wrap them up into a sensible definition of collusion resistance?

In the rest of this section, we will attempt to do just that. Along the way, we will argue that the issues surrounding "collusion" cannot be addressed simply by outlawing some subjectively-chosen set of ill-intentioned actions. Instead, we should design mechanisms that seek consilience by accounting for a broad spectrum of social connections (not just the "bad" ones). Most importantly, we will argue that this approach produces mechanisms that are more pluralistic and closer to optimal. We will then end this section by defining some notation and discussing how one can apply our results to QV.

In the rest of this paper, we will walk through currently-proposed mechanisms, examining potential issues and offering more robust mechanisms to replace them. Section 2 will cover *Pairwise Discounting*. Section 3 will cover *Cluster Match* and two new mechanisms, one of which satisfies our definition of collusion resistance. Section 4 will cover *Offset Match*. Section 5 will briefly describe one promising new mechanism which might combine the strengths of Offset Match and Cluster Match while avoiding the drawbacks of each. Finally, we will conclude in section 6 with a table

summarizing our main results, ideas for future work, and a link to a GitHub repository with code for all of the mechanisms we discuss in this paper.

We hope this work will provide the building blocks for future theoretical and practical work on plural funding. Our analysis of new and old mechanisms in this paper is in the spirit of technical exploration, not because we believe that our new mechanisms are “better”. All proposed mechanisms have strengths and weaknesses, which might lead to them thriving or failing depending on the context of their implementation. Similarly, plenty of other definitions of collusion resistance may be more or less realistic depending on context. Our goal is to paint a broad picture of the landscape of technical possibilities seen from one particular vantage point rather than shoehorn the reader toward a particular mechanism or design philosophy. As research on pluralist mechanism design moves forward, we hope this paper accounts for only a part of an ample space of technical innovation.

1.1 From collusion resistance to rewarding consilience

First, let us dig deeper into the similarities and differences between Sybil agents coordinating via some puppet master and real people coordinating via shared social ties. By Sybil agents, we mean multiple QF accounts controlled by the same entity.

Here it is helpful to ask: why are Sybil agents problematic in the first place? Their existence results in a project getting more funding than it “should” receive, but what do we mean by that? The answer is that when QF awards subsidy funding to a project (as shown in the left and middle diagrams of figure 2), it is doing so to reward agreement among independent agents – but the important word here is *independent*. Sybil agents controlled by a common entity are not independent, so their reward for consilience should be zero.

Seeing things from this angle helps us understand why our case of the software company might raise some red flags while still being qualitatively different from the case with Sybils. The critical thing to notice is that while each employee is a distinct human being (which cannot be said of Sybil agents), there is also one crucial axis along which they are all the same – they are all members of the company. So while we do not want to treat these eight employees as Sybil agents, we might want to award less funding than we would have awarded if those eight contributors had not been cooperating with this company. In other words, since the point of QF is to reward consilience across different agents, given less difference, there is less consilience to reward.

Moreover, besides companies, one could plausibly imagine a multitude of socio-cultural groups (e.g., family, school, sports clubs, online forums, metaverses, et cetera) meaningfully influencing the way members of those groups interact with QF mechanisms. Also, individuals may identify with different groups with different strengths. Therefore, if we want to fully understand all of the ways of conceptualizing socially connected human beings, we need to consider a spectrum of social connectivity, with Sybil agents on one end, normal socially connected individuals in the middle, and the type of selfish, isolated agents that economists like to imagine on the other end. What makes Sybil agents Sybils is that the will of one entity centrally coordinates them. They should be recognized as precisely the same because they all listen to that same entity and that entity alone. To take a rather extreme real-world example, then (and now starting to move rightward down the spectrum), we might think of a Sybil agent as similar to an individual who identifies very strongly with one specific group, and mostly coordinates their actions with the will of that group. However, most people instead belong to many overlapping social groups, each having a particular sway over

that individual's actions, as highlighted by Simmel[17]¹. Of course, the number of such groups an individual belongs to naturally varies and evolves, and some individuals may simply be in fewer groups than others. So as we move to the other end of the spectrum, individuals in fewer and fewer social groups begin to look more and more like the type of self-interested agents that economists usually put into their models – i.e., the *homo economicus*.

Here is the kicker. In theory, QF is the optimal way to distribute capital in a decentralized, self-organizing ecosystem if and only if we assume that all participating agents are selfish rational utility maximizers. In other words, QF treats everyone as if they were at the *homo economicus* end of the above spectrum: it awards extra funding for “cooperation across differences” as if the agents were all as different as they could be and would never think to cooperate based on shared social ties (which are assumed not to exist). So QF is not just sub-optimal when Sybil agents enter the picture: it is not even optimal when human beings enter the picture (see Appendix A for formal proof).

Therefore, adapting QF for “collusion resistance” is not about outlawing a specific type of attack that Sybil agents or even highly coordinated real people can carry out, since such a design philosophy would only acknowledge the *other* extreme end of the above spectrum. Instead, we need to design mechanisms that acknowledge the entirety of the spectrum. Our goal should be to make QF closer to optimal by calibrating the mechanism to give greater rewards for consilience when agents are more different, and at the same time respecting the fact that different agents may lie at different points in a spectrum of social connectedness. At one end of the above spectrum, when agents are identical (Sybils), the mechanism should be able to catch that and completely squash any extra funding. In the middle of the spectrum, when agents are similar to some degree, the system should be able to tamp down its rewards, or else it will overshoot funding levels past what is optimal (again, see Appendix A). Lastly, if agents really are socially atomized, the mechanism should behave like the original proposal for QF. Moreover, the mechanism should be able to account for all of these levels of social connection at once: given an instance with some sybils, some relatively connected agents, and some atomized agents, the mechanism should be able to treat all of them appropriately.

Our long-term goal as researchers should be to craft mechanisms with this type of flexibility, and in this paper we take steps toward that goal. It is essential to prevent Sybil attacks, but if we were to focus exclusively on this, we would wind up with significantly less optimal mechanisms. In any case, many of the most promising approaches to preventing Sybil attacks rely on leveraging social information [10, 3, 11] since Sybil-resistance is very hard to solve in completely trustless settings [5, 13]. So if the tools we need to prevent Sybils already use social information, we could bootstrap even better mechanisms off of that same data. Sybil-resistant and Plural mechanisms may draw on highly similar data: the intersections of social circles. Put in the words of Simmel [18]:

The groups to which an individual belongs form, so to speak, a system of coordinates, in such a way that each newly added group determines the individual more precisely and more unambiguously. The affiliation to one of them still leaves a wide scope for individuality; but the more there are, the less likely it is that other people will show the same group combination, or that these many circles will once again intersect in one point.”

¹In this framework, a person's unique desires independent of any other social ties could be represented as a singleton group containing just that person.

If identity is fundamentally social, it would be a waste to leverage such information to support mechanisms that ultimately ignore sociality; instead, we should harness it to empower socially aware, plural mechanisms.

Before we continue we should note that practically capturing complex social information is non-trivial. An in-depth discussion of this aspect of QF is in Appendix B.

1.2 A heuristic definition of collusion resistance

In this section, we will offer a simple working definition of collusion resistance that attempts to capture all the social possibilities mentioned above. However, before formally defining collusion resistance, it is worth taking a step back and thinking intuitively about how QF doles out capital to projects since this will help us clarify what our formal definition should be capturing.

In particular, we should look at a project's QF income from one agent's perspective. Expanding the power of 2 in the formula for vanilla QF (i.e., the original, individualist formula from (1)) reveals that for any agent i , we can rewrite the funding amount as

$$c_i + \left(\sqrt{c_i} \cdot 2 \cdot \sum_{j \neq i} \sqrt{c_j} \right) + \left(2 \cdot \sum_{j, k \neq i} \sqrt{c_j} \cdot \sqrt{c_k} \right). \quad (2)$$

From agent i 's perspective, the funding amount consists of their contribution plus the root of their contribution times some constant plus another constant.

Agent i might wonder what difference they made by participating in the mechanism rather than directly contributing to the project through another payment channel. If agent i had chosen this route, then the project would have still received c_i (via agent i 's direct contribution) and $\sum_{j, k \neq i} \sqrt{c_j} \cdot \sqrt{c_k}$ (via the other agents who did participate in QF). However, the middle term would disappear in (2). Thus, by participating in QF, agent i brought an extra

$$\sqrt{c_i} \cdot 2 \cdot \sum_{j \neq i} \sqrt{c_j} \quad (3)$$

of subsidy funding to the project. Notably, this quantity is $O(\sqrt{c_i})$. In other words, as agent i contributes more and more (while holding others' contributions constant), the subsidy amount given by the system grows at a rate of $\sqrt{c_i}$ times some constant.

This phenomenon is illustrated visually in figure 3, which shows a 3-person QF funding diagram "seen" from the perspective of agent 3. For them, the green area is their contribution, and the grey areas are the amount that QF would award to the project regardless of their contribution. The yellow rectangles comprise the portion of the subsidy amount that QF awards to the project due to agent 3's participation in the mechanism (holding other contributions constant). Note that one side of each yellow rectangle has length $\sqrt{c_i}$. So as c_i increases, the yellow rectangles grow in size at a rate proportional to $\sqrt{c_i}$.

The fact that a contribution of c_i provokes an extra $O(\sqrt{c_i})$ of subsidy funding from vanilla QF is crucial since it means that there are diminishing returns on the amount of subsidy capital any one agent can get out of the system. These diminishing returns on individual inputs are the core property that we will want to retain when designing collusion-resistant QF mechanisms. Just as any one agent should see decreasing returns on the "extra capital" provoked by their contribution



Figure 3: QF with three contributors from agent 3’s perspective. Gray areas are out of agent 3’s control. The green area is agent 3’s contribution. The yellow areas are the portion of the subsidy amount that QF awards because of agent 3’s contribution (keeping everything else constant).

(i.e., the size of the yellow areas in figure 3), any group of agents (be they Sybil agents controlled by one person, or individual people coordinating via social ties) should see decreasing returns on their collective contribution, in the limit. In this work, we will call it collusion whenever groups or individuals can achieve linear (or super-linear) growth in the subsidy doled out by QF as either their collective or individual contribution grows or as they add more agents to the group. This informal definition will suffice to understand most of the arguments in the paper, but we offer a more formal definition in Appendix D in order to rigorously prove that one of our mechanisms is collusion resistant (Theorem 1, proved in Appendix E).

We should be clear about some potential shortcomings of our definition. First, our definition hinges on the asymptotic properties of the funding mechanism. However, the real world does not necessarily bear out asymptotic behavior – it may not be realistic to expect agents or groups to push the mechanism by contributing asymptotically large amounts of money, even if an asymptotic vulnerability did exist. Second, although we consider the effect of new agents on the funding amount, we do not consider the effect of adding new groups because we assume that system administrators will set policies to ensure only legitimate groups participate in the system. Lastly, this is a “worst case” definition in which we consider all groups equally likely to collude (or, on the other side of the coin, equally deserving of funding attenuation in the name of consilience). Different groups may have differently strong social fabrics in practice. Indeed, we hope future work might focus on finding more comprehensive definitions and understandings of social groups beyond binary membership rosters. This work will likely look more closely at nuanced differences in how individuals relate to different groups and at nuances in how different groups may relate to each other.

Nevertheless, this definition will help us illuminate some important properties of QF mechanisms, which is the primary goal of this paper.

1.3 Notation

In order for our mechanisms to satisfy our anti-collusion property concerning social groups (or groups of Sybil agents), we will assume that there is some distribution of identity certificates about group memberships that legitimately conforms to the rules of the QF operator (in other words, we assume accurate information about relevant social groups).

Let us introduce some notation. Let $N = \{1, \dots, n\}$ be a set of agents and let G be a bag of groups². Each group $g \in G$ is a non-empty subset of N . Let T_i denote the groups to which agent i belongs. We will sometimes refer to G as the “group structure”.

In this paper, we can think about one project at a time, so we do not need any notation for differentiating projects. Let c_i denote the contribution of agent i to a project, and let c be the vector of all contributions. For a group $g \in G$, let c_g denote the combined contributions of every group member, i.e., $c_g = \sum_{i \in G} c_i$.

In this paper, we will talk about “QF formulas”. A QF formula is a function that takes in contribution and group membership information and returns a funding amount. We have already discussed one QF formula: the formula for “vanilla QF”, described in (1).

Lastly, when doing algebra, we will sometimes use $O(1)$ and $O(\sqrt{c_i})$ to denote generic terms that are constant with respect to c_i and grow at a rate of $\sqrt{c_i}$ asymptotically, respectively.

1.4 Collusion-resistant QV

Before continuing to the collusion-resistant QF formulas, we will take a brief detour to note that the modifications to QF discussed in this paper also directly apply to make QV more pluralistic. QV sees agents spend “voice credits” on an issue, which it converts to effective votes at quadratic costs. So if agent i spends v_i voice credits towards an issue, then the effective number of votes given to that issue is

$$\sum_{i \in N} \sqrt{v_i}. \quad (4)$$

Notice that if we let the v_i values denote contributions (i.e., letting $c_i = v_i$), (4) can be re-written as the root of the total amount of funding under vanilla QF:

$$\sum_{i \in N} \sqrt{v_i} = \sqrt{\left(\sum_{i \in N} \sqrt{c_i} \right)^2}. \quad (5)$$

At first glance, this might seem like a needlessly complicated way to re-write the formula for QV. However, the point is that instead of putting the formula for vanilla QF inside the square root on the RHS of (5), we could plug in a collusion-resistant formula instead. In other words, this way of generalizing QV (as the root of the funding amount given by *some* QF formula) opens the door to experiments with more pluralistic quadratic voting.

²A bag is a set in which duplicate (or triplicate, et cetera) elements are allowed. We use a bag instead of a set to allow to allow multiple groups with the same members.

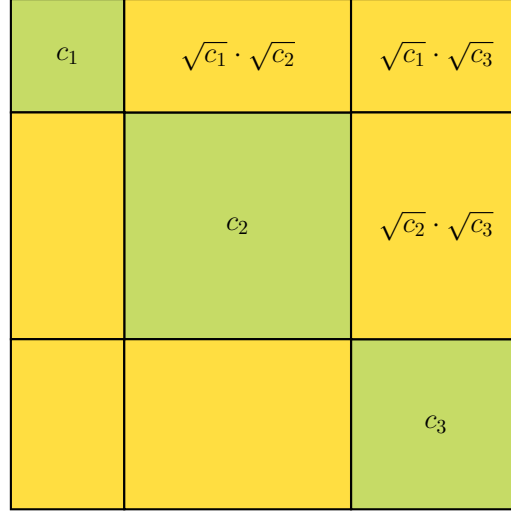


Figure 4: The quadratic funding diagram from figure 1, with the yellow area split into rectangles. Notice that each rectangle's area is $\sqrt{c_i} \cdot \sqrt{c_j}$ for some $i \neq j$.

We refer to each yellow rectangle as an *interaction term*.

2 Pairwise Discounting

Buterin first suggested a form of Pairwise QF and Discounting in [1]. Returning to our geometric interpretation of QF will help us understand this mechanism. Figure 4 shows an example QF funding diagram, with the subsidy amount (yellow area) split into rectangles. The critical observation is that each yellow rectangle has area $\sqrt{c_i} \cdot \sqrt{c_j}$ for two different agents i and j (and two such rectangles exist for every distinct pair of agents). Call the area of each pair of yellow rectangles the *interaction term* for agents i and j . We can think of each interaction term as the portion of the subsidy that the mechanism rewards based on i and j both contributing to the project (i.e., the reward for cooperation across differences, local to just i and j). The main idea behind Pairwise Discounting is that we should scale down the areas of these yellow rectangles according to how similar any two agents are.

Formally, this means introducing coefficients $k_{i,j}$ for each pair of agents i, j , with each $k_{i,j}$ being closer to 0 if i and j have a more similar social background. Inversely, $k_{i,j}$ should be closer to 1 if i and j have more diverse social backgrounds (with the exception that $k_{i,i} = 1$ for all i). These coefficients can be calculated based on information about groups or from information about the agents' contribution amounts to the suite of projects up for funding. For example, one of us proposed [6] to define the coefficients as

$$k_{i,j} = 1 - \frac{|T_i \cap T_j|}{|T_i \cup T_j|}. \quad (6)$$

Given some way of calculating $k_{i,j}$'s, the total funding amount under Pairwise Discounting is

$$\sum_{i \in N} c_i + \sum_{i \in N} \sum_{j \neq i \in N} k_{i,j} \cdot \sqrt{c_i} \cdot \sqrt{c_j} \quad (7)$$

Note that with all $k_{i,j} = 1$ this simplifies to the amount prescribed by vanilla QF.

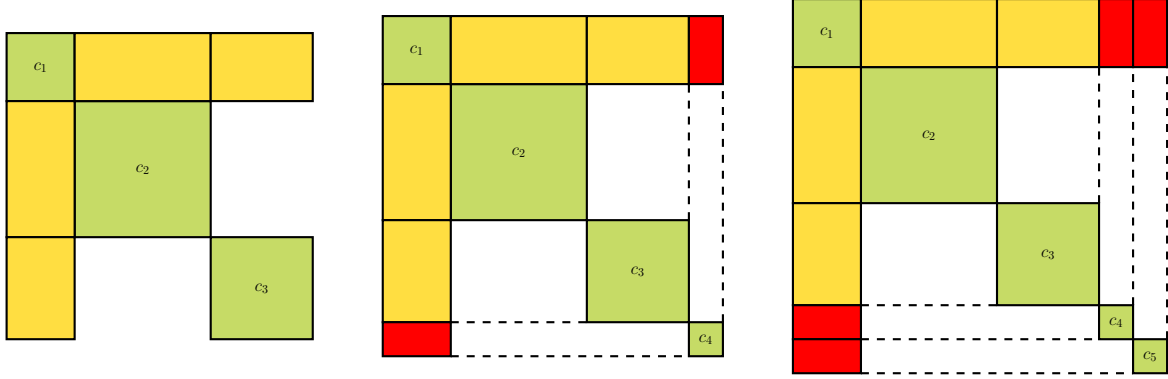


Figure 5: Pairwise Discounting with colluding agents 2, 3, 4... and non-colluding agent 1. Removing the interaction terms for every pair of colluding agents, the colluding party can still eke out a constant amount of subsidy capital (represented by each pair of red rectangles) for every unit spent, by giving each new unit via a new agent's account.

2.1 Pairwise Discounting is vulnerable to collusion via many agents

There is a sense in which Pairwise Discounting is vulnerable to collusion no matter how one calculates the $k_{i,j}$ coefficients. In the following example, let us assume that the system can perfectly tell when two agents collude or have a broad shared socio-cultural background. We will show that the agents can nevertheless exploit the system if 1) at least one non-colluding agent happens to also contribute to the project and 2) the colluding party can recruit extra agents to contribute to the project.

Imagine a project with three contributors. Assume that agents 2 and 3 collude, and the system sets $k_{2,3} = 0$. The leftmost diagram in figure 5 shows such a scenario. However, note that if agents 2 and 3 recruit agent 4 to give just \$1 of funding, the subsidy amount will still go up by $\$2\sqrt{c_1}$ (represented as red rectangles in the middle diagram of figure 5), even if the system can tell that agents 2,3 and 5 are all in cahoots. Indeed, as long as the colluding agents can keep evoking contributions from new accounts, they can keep gaining a constant amount ($\$2\sqrt{c_1}$) of subsidy capital for each new unit spent – even if we set $k_{i,j} = 0$ for *all* pairs of colluding i and j .

In other words, letting c_g be the combined contributions of agents 2,3,4,..., these agents can collectively eke out $O(c_g)$ in subsidy funding from the mechanism. This phenomenon breaks our requirement that as any group adds more members (who only join that group), they see diminishing returns on the matching funds doled out to a project.

So, calculating collusion coefficients based on the similarities between pairs of agents is insufficient to guarantee collusion resistance, at least under the assumption that colluding groups may recruit extra agents. The following mechanism, Cluster Match, gets around this vulnerability presented by looking at information about groups of agents rather than pairwise comparisons.

3 Cluster Match

Cluster Match was proposed in [20]. The idea is to put agents in the same group “under the same square root”. In other words, to take the above example, if agents 2,3,4,... were in a colluding group (and no other groups existed), we would calculate the total funding amount as

$(\sqrt{c_i} + \sqrt{c_1 + c_2 + \dots})^2$. This method fixes the vulnerability presented above – no matter how many agents the colluding group recruits, they still see diminishing returns on their collective contribution.

However, there are still some details to iron out since, in general, people can be in many social groups at once. The idea in the original proposal from [20] went as follows: let G be a bag of groups as defined in section 1.3. Then the original proposition was for Cluster Match to award

$$\left(\sum_{g \in G} \sqrt{\sum_{i \in g} \frac{c_i}{|T_i|}} \right)^2 \quad (8)$$

in funding. One way of interpreting this formula is to assume that when an individual contributes, they do it partially on behalf of each group to which they belong. So we could split this agent's contribution evenly among their groups. After doing this calculation for all agents and summing, we are then able to say that it is “as if” each *group* had contributed a certain amount (specifically, this amount is $\sum_{i \in g} \frac{c_i}{|T_i|}$). So then we do vanilla QF but on the group contributions that we have constructed.

Notably, one could also imagine uneven contribution splits based on weights that signify which groups have more or less meaning to the agent. In that case, an agent might have different fractional weights for different groups. One would need to require that each weight is between 0 and 1 and that the weights collectively sum to 1. However, here we assume an even split for simplicity.

The issue with Cluster Match is that an individual i can get $O(c_i)$ subsidy capital out of the system whenever they are in more than one group. It is easiest to see this if we consider a project with one contributor in two groups, X and Y. The left side of 6 shows a visual interpretation of the funding amount – here, it is easy to see that QF will double the capital contributed by this lone agent. With only one agent contributing, the mechanism should probably not award any subsidy, much less a linearly growing subsidy.

To be more concrete, we can show that whenever $|T_i| > 1$, agent i can get $O(c_i)$ in subsidy funds out of vanilla Cluster Match. That's because for any agent i , we have

$$\left(\sum_{g \in G} \sqrt{\sum_{j \in g} \frac{c_j}{|T_j|}} \right)^2 \geq \left(\sum_{g \in T_i} \sqrt{\frac{c_i}{|T_i|}} \right)^2 = \left(|T_i| \cdot \sqrt{\frac{c_i}{|T_i|}} \right)^2 = |T_i|^2 \cdot \frac{c_i}{|T_i|} = |T_i| \cdot c_i. \quad (9)$$

9 shows that the total funding amount awarded under vanilla Cluster Match is at least $c_i \cdot |T_i|$ for any agent i , so whenever $|T_i| > 1$, agent i can contribute c_i and see the total funding amount increase by at least $2c_i$. As one can see, we would have an $O(c_i)$ growth in matching funds here, so Cluster Match is not collusion-resistant.

3.1 Squared Cluster Match

We can remedy this issue by tweaking the formula slightly. Call this new mechanism “Squared Cluster Match”: it awards total funding amount

$$\left(\sum_{g \in G} \sqrt{\sum_{i \in g} \frac{c_i}{|T_i|^2}} \right)^2. \quad (10)$$

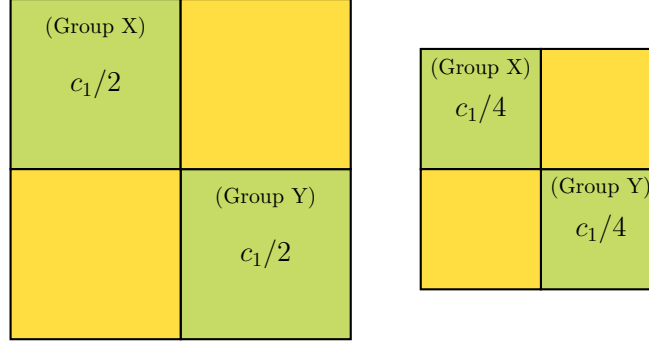


Figure 6: Left: a QF diagram showing funding under the initially proposed Cluster Match with one contributor in two groups (note that the total funding amount is $2c_1$). Right: Squared Cluster Match with one contributor in two groups (note that the total funding amount is c_1).

Going back to our simple one-person example (shown visually on the right side of figure 6), we can now see that the total funding amount awarded by the mechanism, in this case, is just c_i . I.e., the lone agent sees no extra capital contributed to the project.

For a more general treatment of how Squared Cluster Match alleviates this problem, we can show how the math in 9 changes under the formula for squared Cluster Match. For a group g , let $r_g = \sum_{j \in g \setminus \{i\}} c_j / |T_j|$. Then

$$\left(\sum_{g \in G} \sqrt{\sum_{j \in g} \frac{c_j}{|T_j|^2}} \right)^2 = O(1) + \sum_{g \in T_i} \sum_{h \in T_i} \sqrt{r_g + \frac{c_i}{|T_i|^2}} \cdot \sqrt{r_h + \frac{c_i}{|T_i|^2}} \quad (11)$$

$$= O(1) + |T_i|^2 \cdot \left(\frac{c_i}{|T_i|^2} + O(\sqrt{c_i}) \right) = c_i + O(\sqrt{c_i}) + O(1) \quad (12)$$

where the first equality comes from expanding the power of 2 and noting that all interaction terms involving groups outside of T_i are constant with respect to c_i (recall that an *interaction term* is the portion of the subsidy funding calculated by multiplying the square roots of two entities' contribution, and is explained visually in Figure 4). So, the total funding amount under Squared Cluster Match will be c_i plus some constant $O(\sqrt{c_i})$ terms. So the subsidy amount seen by agent i (total funding minus c_i) will not grow linearly, satisfying at least one aspect of our definition of collusion resistance.

However, while individuals can not exploit Squared Cluster Match, groups of agents can. In the next section, we will explain this vulnerability in more detail and present another tweak to Cluster Match that ameliorates the issue.

3.2 Connection-Oriented Cluster Match

To motivate the next mechanism, let us consider a slightly more complicated example with three different agents. Suppose agents 1 and 2 are in group X and agents 2 and 3 are in group Y. The diagram on the left side of figure 7 shows an illustrative visual representation of Cluster Match under these conditions. The problem is that under these conditions, group X can retrieve a linearly growing amount of matching funds as they increase their collective contribution (the same is valid for group Y symmetrically, but we will focus on group X here). Let c_X be the total amount

		(Group A)		
		$\frac{c_1}{2}$		$\sqrt{\frac{c_1}{2}} \cdot \sqrt{\frac{c_2}{2}}$
(Group X)			(Group B)	
$\frac{c_1}{2} + c_2$	$\sqrt{\frac{c_1}{2} + c_2} \cdot \sqrt{\frac{c_1}{2} + c_3}$		$\frac{c_1}{2} + \frac{c_2}{2}$	
	(Group Y)			(Group C)
	$\frac{c_1}{2} + c_3$			$\frac{c_2}{2}$

Figure 7: Two illustrative examples of vanilla Cluster Match, with the areas (funding amounts) of some interaction terms highlighted. In the leftmost example, group X can achieve linear growth in the subsidy amount (relative to their contribution) by exploiting the interaction term for group X and group Y . This exploit is possible because groups X and Y share members. However, the rightmost example shows that more than scaling down the interaction terms of groups that share members is needed to guarantee collusion resistance. In the rightmost example, groups A and C do not share members, but the members of A and C are in another group (group B). Therefore, even if one removed all other interaction terms, group B would still achieve a linear growth in the subsidy amount by exploiting the interaction term for groups A and C .

contributed by members of the group X (so $c_X = c_1 + c_2$). For simplicity, assume $c_1 = c_2 = c_X/2$ (the results we obtain do not depend on exactly how c_1 and c_2 relate to each other). Then we can write each yellow rectangle's area as

$$\sqrt{\frac{c_1}{2} + c_2} \cdot \sqrt{\frac{c_1}{2} + c_3} = \sqrt{\frac{3c_X}{4}} \cdot \sqrt{\frac{c_X}{4} + c_3} = \sqrt{\frac{3c_X^2}{16} + \frac{3c_X c_3}{4}}. \quad (13)$$

The vital thing to notice here is that the first term inside the rightmost square root of (13) is c_X^2 multiplied by some constant. So once we take the square root, we will have a linear expression in c_X . In other words, as group X raises its contribution, they see an $O(c_X)$ increase in matching funds. Squared Cluster Match is also vulnerable to this type of exploit, although the picture is more complex than in vanilla Cluster Match – a full explanation is in Appendix C.

In the above example, note that the linear subsidy growth enjoyed by group X came from the interaction term for groups X and Y and was due to the fact that groups X and Y shared members. One first pass at ameliorating this issue might involve somehow attenuating the interaction terms for groups that share members – say, by removing all individuals in both groups from the calculation. However, this won't suffice to solve the problem. The rightmost diagram of figure 7 shows why.

In the rightmost diagram of figure 7, we have two agents and groups A , B , and C . Agent 1 is in groups A and B , and agent 2 is in groups B and C . Similarly to above, let $c_B := c_1 + c_2$ and $c_1 = c_2 = c_B/2$. Now, we can write the interaction term for groups A and C as

$$\sqrt{\frac{c_1}{2}} \cdot \sqrt{\frac{c_2}{2}} = \sqrt{\frac{c_B}{4}} \cdot \sqrt{\frac{c_B}{4}} = \frac{c_B}{4} \quad (14)$$

which is linear in c_B . So here we have subsidy growth linear in a group's contribution occurring in the interaction term for groups A and C , but it is not because groups A and C share members. Instead, a different group (B) was able to exploit the A, C interaction term to draw out linear growth in the subsidy relative to *that* group's contribution. Simply keeping an eye on which pairs of groups share members, and attenuating the interaction terms for those pairs of groups, is not enough to guarantee collusion resistance.

One path to collusion resistance that we have found involves taking a broader look at the social connections illustrated by group memberships. The idea is to attenuate the contributions of agents within interaction terms when they are socially connected to the group on the “other side” of the interaction term, for a relatively broad definition of social connectivity. When calculating group X 's side of the interaction term for groups X and Y , we will still sum over the contributions of all agents in group X , but we will attenuate each agent's contribution to various degrees within that sum. If an agent is also in group Y or they are in some shared group with some member of Y , we will take the square root of their contribution within the sum. If neither of these things are true, we will not attenuate the agent's contribution.

One way of technically specifying this is to define a function K which takes in an agent i and a group h and returns some attenuation of c_i . Formally,

$$K(i, h) := \begin{cases} \sqrt{c_i} & \text{if } i \in h \text{ or } \exists j \in h \text{ such that } |T_i \cap T_j| \geq 1 \\ c_i & \text{otherwise} \end{cases} \quad (15)$$

Now we can write the QF formula for a new mechanism, *Connection-Oriented Cluster Match*, as

$$\sum_{g \in G} \sum_{i \in g} \frac{c_i}{|T_i|} + \sum_{g \in G} \sum_{h \neq g \in G} \sqrt{\sum_{i \in g} \frac{K(i, h)}{|T_i|}} \cdot \sqrt{\sum_{j \in h} \frac{K(j, g)}{|T_j|}} \quad (16)$$

There are a few things to observe about this formula. First, note that the leftmost double-sum of 16 adds all the group contributions (i.e., the green squares in our diagrams). The sum over all distinct pairs of groups calculates the interaction terms. These interaction terms look as they would under vanilla Cluster Match, except we attenuate contributions according to $K(i, h)$ or $K(j, g)$. If these $K()$ terms all evaluated to the contributions of their respective agents (e.g., if all groups had distinct members), we would recover vanilla Cluster Match.

In order to rigorously prove that Connection-Oriented Cluster Match is collusion resistant, it is necessary to lay out a more rigorous definition of collusion resistance: we do this in Appendix D. With this definition in hand, we have the following theorem:

Theorem 1. *Connection-Oriented Cluster Match satisfies the definition of collusion resistance in Appendix D.*

This theorem is proved in appendix E.

We can also argue for the sensibility of Connection-Oriented Cluster Match by appealing to the intuitive foundations of QF. Intuitively, we want a mechanism to reward a project whenever two distinct agents like it. This idea should also apply for groups; we should reward projects liked by distinct groups. However, if two groups consist mainly of the same agents (or if most members of both groups are all in another third group), how distinct are they after all? In Connection-Oriented Cluster Match, we try to reduce the reward for “agreement across differences” between pairs of groups that are not as different.

There is something relatively coarse about Connection-Oriented Cluster Match since every contribution is either left alone or square rooted inside interaction terms. One could imagine the function $K(i, h)$ returning an attenuation of c_i much more closely tuned to the specifics of the social situation. For example, c_i could be attenuated more if the agent is actually in the other group or attenuated less if there are very few social connections between the two groups. Similarly, one could imagine the mechanism using a finer-grained measure of *social distance* between two agents. For example, one could imagine a mechanism that understands when two agents are not in common groups but have a multi-hop connection of some length. The present coarseness of the mechanism is due to the simplified model we are working in – using $\sqrt{c_i}$ as the only attenuation option is the easiest way to prove collusion resistance under our definition. Similarly, keeping track of multi-hop social connections is not necessary to prove collusion resistance. However, the space of options for tuning a mechanism like this may be much more extensive.

There is some reason to believe that the mechanism we present in section 5, *Eigen Match*, addresses these issues of coarseness.

4 Offset Match

Offset Match was proposed in [20]. The basic idea is to scale down the contributions of agents according to their social centrality, thereby increasing the effect that more unique agents have on the funding amount. In this paper, Offset Match is a dark horse: the issues we will discuss (and

solutions we offer) will be of a different flavor than the rest of the paper. Nevertheless, before we get to those, let us define the mechanism.

First, define the “correlation score” between any two (ordered) agents as

$$s_{i,j} = \frac{|T_i \cap T_j|}{|T_i|}. \quad (17)$$

These correlation scores attempt to capture how much agent i internalizes j ’s utility or how much i “cares about” j . With these correlation scores in hand, the next step is to find coefficients α_i solving the system of equations

$$\sum_{i \in N} \alpha_i \cdot s_{i,j} = 1 \quad \forall j \in N. \quad (18)$$

One intuitive explanation for this system is that we are trying to find coefficients to scale correlation scores so that everyone is “cared about” equally strongly. Once we have these α_i values, the funding amount prescribed by Cluster Match is

$$\left(\sum_{i \in N} \sqrt{\alpha_i \cdot c_i} \right)^2. \quad (19)$$

4.1 When is the system solvable?

The first important thing to notice about Cluster Match is that calculating the α_i coefficients requires solving a system of linear equations. However, this system may not have a solution. For example, $N = \{1, 2, 3\}$ and $G = \{\{1, 2\}, \{2, 3\}\}$ yields the coefficients $s_{1,2} = s_{3,2} = 1$, $s_{2,1} = s_{2,3} = 1/2$, and $s_{1,3} = s_{3,1} = 0$. But the system

$$\alpha_1 + \frac{\alpha_2}{2} = 1 \quad (20)$$

$$\alpha_1 + \alpha_2 + \alpha_3 = 1 \quad (21)$$

$$\frac{\alpha_2}{2} + \alpha_3 = 1 \quad (22)$$

has no solution since requiring the first and last equalities means that $\alpha_1 + \alpha_2 + \alpha_3 = 2$. Therefore, characterizing where this system of equations is solvable is an important first step toward understanding Offset Match’s practical usability. We offer one such characterization, as described by the following Theorem.

Theorem 2. *Let agent i ’s “group profile vector” be a vector $(1_{i \in g})_{g \in G}$ where $1_{i \in g}$ is an indicator variable that is 1 if $i \in g$ and 0 otherwise. Then Offset Match is feasible (i.e., the system of equations has a solution) if and only if there is no linear dependence among the group profile vectors.*

We prove this theorem in Appendix F.1. Two corollaries follow (again, with proofs in the appendix). First, if the number of groups is smaller than the number of agents, we are guaranteed to wind up with an unsolvable system of linear equations, rendering Offset Match useless. However, the second corollary offers good news. Assigning each agent to a new singleton group containing just that agent, we are guaranteed to get linearly independent group profile vectors; therefore, Offset Match is always feasible if we undergo this pre-processing step.

4.2 Offset Match and individual rationality (IR)

So far, we have yet to mention a vital property to consider when designing QF mechanisms: individual rationality (IR). More broadly, in mechanism design literature, requiring IR means that agents' outcomes should never be made worse by participating in the mechanism. If we did not care about IR, designing a collusion-resistant mechanism would be easy: we could ignore all individual contributions and group information and return a final funding amount of zero for every project, no matter what. This mechanism prevents collusion since no individual or group of agents could use it to extract linearly growing subsidy funds. However, the problem is that no one would use it in the first place. Instead of contributing to a mechanism that just swallowed their capital and did nothing in return, agents might contribute via some other channel or not contribute anywhere at all – so the above exemplary mechanism is not individually rational. More formally, in our context, requiring IR means that if an individual contributes c_i , the *total funding amount* should rise by at least c_i .

Offset Match doesn't always satisfy IR. Characterizing exactly when Offset Match is IR and when it is not is an important area for future work, but for now, we note that one can construct an n -agent example where all but one agent has $\alpha_i = 0$ (the last agent has $\alpha_n = 1$). In other words, there exist cases where almost nobody has the incentive to use the Offset Match mechanism. Our construction details are in Appendix F.2.

It is not clear whether every IR is necessary in every setting. The property is likely critical when people use universal capital (like US dollars, bitcoin, or time) and have plenty of alternative channels to allocate that capital to. However, it might be less important in other cases. For example, a community might run QF via some currency dedicated to that community and mechanism. In this case, since agents could not use that specific currency for anything else, IR might be less of a concern.

5 Eigen Match

Before concluding, we will briefly present one more idea for a mechanism. This idea is relatively undeveloped but might be one of the most promising avenues for future work.

The idea behind this mechanism, which we call *EigenMatch*, is to attempt to combine the virtues of both offset and cluster match while avoiding the pitfalls of each. It is a Cluster-Match-like procedure but under a more generalized notion of a “group”. Instead of taking in a set of groups, this mechanism would work directly from a social graph.

The mechanism starts by calculating the eigenvectors of the adjacency matrix of the social graph. Then, one takes the set of all eigenvectors as the set of groups and includes each agent in each group with a weight corresponding to that agent's index of that group's eigenvector. Then, one utilizes these weights to calculate “group contribution” and does normal QF on the group contributions, as in Cluster Match.

Formally, let E be the set of eigenvectors of a social graph. Then EigenMatch might calculate the funding amount as

$$\left(\sum_{\mathbf{v} \in E} \sqrt{\mathbf{v} \cdot \mathbf{c}} \right)^2 \quad (23)$$

where c denotes the vector of all contributions. Note that this is just a sketch of the idea. The weights of each agent in each group do not necessarily need to be *exactly* that agent’s weight in that group’s eigenvector. Moreover, one could consider somehow using each eigenvector’s eigenvalue in the calculation.

Like Offset Match, this mechanism incorporates all the agent-to-agent pairwise social information contained in an n by n matrix, rather than just forming local clusters. But like Cluster Match, it is about clustering like people rather than just offsetting effects that emerge from coordination (which can potentially violate IR). We hope that further exploration of this mechanism will improve understanding of plural QF.

6 Conclusion

We hope this paper has clarified some issues with currently proposed collusion-resistant QF mechanisms and offered better alternatives. Although we found that the already-proposed Cluster Match, Pairwise Discounting, and Offset Match were all prone to various social and technical issues, we presented several new mechanisms (Squared Cluster Match, Connection-Oriented Cluster Match, and EigenMatch) which build off of the already-proposed mechanisms in promising ways. Additionally, one new mechanism – Connection-Oriented Cluster Match – is entirely collusion resistant. On the other hand, we stress that the strengths and drawbacks we have pointed out only matter relative to the particular technical framework we have chosen for this paper. Since there are countless other technical perspectives one could take, one should not take our results as a treatise on which mechanisms are “good” and which are “bad”, but instead as simply providing a deeper technical understanding of plural QF. One can find a summary of the results we have explored from our particular technical standpoint in table 1. Aside from the results listed in the table, we have shown that semi-coordinated agents break QF’s optimality guarantee (appendix A); characterized the settings under which Offset Match is feasible (appendix F.1); and shown that one can use all of these results to build collusion-resistant QV mechanisms as well (section 1.4). Lastly, we have coded up all of these mechanisms and put them in a GitHub repository³ so that others can explore them and run numerical experiments.

³<https://github.com/Jmiller4/plural-qf>

Table 1: Summary of Mechanisms and Results

<i>Mechanism</i>	<i>Key Idea</i>	<i>Exploitable?</i>	<i>Individually Rational?</i>
Pairwise Dis-counting (section 2)	Attenuate “interaction terms” between pairs of similar agents.	Yes, groups can exploit by adding new members (section 2.1)	Yes
Cluster Match (section 3)	Compile contributions into different group contribution amounts, then do QF on those group contributions.	Yes, individuals in more than one group and groups with shared members can exploit (section 3)	Yes
Squared Cluster Match (section 3.1)	Like Cluster Match, but individuals contribute $c_i/ T_i ^2$ to each group they’re in.	Depending on exact contribution amounts, groups with shared members can exploit (appendix C)	Not always, depending on exact contribution amounts (appendix C)
Connection-Oriented Cluster Match (section 3.2)	Like Cluster Match, but attenuate interaction terms between pairs of similar groups.	No: collusion resistance is proved in Appendix E	Yes
Offset Match (section 4)	Calculate coefficients to estimate social centrality, scale down individual contributions by those coefficients, then do QF on scaled contributions.	Unclear	Not always: exist scenarios where almost all agents have their contribution erased (appendix F.2)
Eigen Match (section 5)	Like vanilla cluster Match, but use a set of pseudo-groups based on the eigenvectors of a social graph.	Unclear	Unclear

We see several important avenues for future work, many of which bleed into each other. As noted in section B, it is unlikely that any mechanism could capture the full scope of social connections experienced by individuals. Therefore, it is vital to somehow estimate how distortive or harmful a plural mechanism could be in the absence of complete information. Second, we define groups and group membership in a very simple way: for us, groups are just sets and every member of a group relates to it in the same way. Research that tries to explore variance among group members, the way different groups relate to each other, and other more nuanced aspects of groups could be invaluable. Third, although we were able to show that Connection-Oriented Cluster Match is collusion-resistant, there is plenty of space to fine-tune the mechanism to make it less coarse and more practically usable. Fourth, there is much we still don’t know about Squared Cluster Match, Eigen Match, and Offset Match. Lastly, we are quite curious about whether there exists some deeper theory of how socially-connected individuals make strategic decisions which can help us understand what we fundamentally mean when we talk about consilience.

On a broader scale, we also hope that this work contributes to the discussion on what economic thought can look like when we break out of the traditional framework that sees agents as selfish and isolated. Of course, there are areas of life where agents are more likely to act like *homo economicus*, but these areas do not represent the totality of human behavior. For example, ethnographic and economic fieldwork has shown time and time again that members of communities often behave in a way that is logical but not wholly selfish (e.g., [15, 14]). If we are going to build tools to help these communities, we had better develop ways of thinking about how they work and incorporating those insights into what we build. Hopefully, this work can contribute to that conversation and help drive a sustainable pluralist future.

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A Naturally “semi-coordinated” agents break QF’s optimality

Here we will show a simple model of partial coordination in which QF fails to be optimal and then discuss its implications.

A.1 Model and proof of sub-optimality

Consider a world with just one project and $n > 1$ agents. Suppose that agent i has the utility function $V_i(F)$ where F is the level of funding to that project. Let $V(F) = \sum_i V_i(F)$ be the society-wide utility for funding level F . As in [2], suppose that all individual valuation functions are concave, smooth, and increasing (therefore V is too). [2] showed that QF chooses F such that $V'(F) = 1$.

However, this ceases to be true if we imagine that agents can consider the utility of others when contributing. We imagine that the set of agents does not explicitly plan how to coordinate their contributions but instead tweak their contributions based on a consideration of others. We can model this by having agent i using the function $\hat{V}_i(F) = v_i(F) + \beta \sum_{j \neq i} v_j(F)$ (for some $\beta > 0$) instead of $V_i(F)$ when deciding on a contribution amount. Here β can be thought of as the degree to which agents internalize each others’ utilities. The idea is that instead of contributing to maximize V_i , agents will contribute to maximize $\hat{V}_i$.

Theorem 3. *If agents contribute to maximize $\hat{V}_i$, vanilla QF does not maximize utilitarian social welfare.*

Proof. Following the math in section 4.1 of [2], their contribution will be chosen to maximize

$$\hat{V}_i \left(\left(\sum_j \sqrt{c_j} \right)^2 \right) - c_i \quad (24)$$

which, again following 4.1 of [2], will have to satisfy

$$\frac{\hat{V}'_i(F) \left(\sum_j \sqrt{c_j} \right)}{2\sqrt{c_i}} = 1 \Leftrightarrow \hat{V}'_i(F) = \frac{\sqrt{c_i}}{\sum_j \sqrt{c_j}} \quad (25)$$

by differentiation. Then summing across agents yields

$$\sum_j \hat{V}'_j(F) = 1 \Leftrightarrow (1 + (n-1)\beta)V'(F) = 1 \Leftrightarrow V'(F) = \frac{1}{1 + (n-1)\beta}. \quad (26)$$

So QF has chosen a level of funding F such that $V'(F) < 1$. Since V is concave, smooth, and increasing, its slope is 1 before it is any constant less than 1. In other words, we have overshot our optimal funding level, giving extra units of funding to the project that society would have been better off saving. $\square$

A.2 Discussion

For simplicity, we assumed in this proof that all agents internalize all other agents' utilities symmetrically. However, note that as long as *some* agent internalizes at least one other agent's utility to some degree, the above proof will still go through since the leftmost expression in (26) will still be greater than $V'(F)$.

However, a limitation of this proof is that despite some effort, we could not tie its modeling assumptions directly in to our definition of collusion resistance. Here, it only illustrates how naturally coordinating agents lead to sub-optimal funding levels under QF.

B Practical considerations for capturing context mechanistically

In practice, it is not easy to find the right identity policies for democratic processes like QV and QF, i.e., defining what social circles get to be recognized and participate in these mechanisms. Friedrich Schiller's 1797 question in *Die Kraniche des Ibykus*, "Who counts the peoples, calls the names?" seems just as relevant to the Plurality paradigm today. We believe that to optimally deploy social technologies like QF, one needs to satisfy the following properties.

- One needs to verify specific attributes about the participants that ensure the participants are affected by or knowledgeable about the questions of the democratic process. Otherwise, the process might lead to minority or local group oppression, disenfranchise essential individuals from the decision, or both phenomena[21].
- Further, one needs to recognize participants' sociocultural attributes to increase cooperation across difference. These measures are specifically critical to address collusion resistance, as defined in the section below.

Since every collective decision comes with a unique context, it is questionable whether one can find universal solutions to these problems. The questions about "who should participate" and "which groups we should promote consilience between" depend on the substance of the funding decisions at hand. This substance is at least as much sociological as it is game-theoretical. For example, in a QV within a firm, one might want to foster consilience between different departments explicitly. In contrast, in a QV about urban infrastructure, the departments of that particular firm might be significantly less relevant (if at all) than consilience between citizens from different neighborhoods. The point we want to illustrate here is that there may hardly be a globally valid group mapping equally applicable to diverse decision-making processes. Instead, defining these parameters in a specific context will require exploration across a range of disciplines, including, for example, sociology, economics, statistics, and the opinions of the associated polity.

Because it seems unlikely that any one institution could ever understand the entire web of social connections or the full context of a democratic decision, a critical area for future work is better understanding how plural QF behaves in the absence of complete information, or in the presence of incorrect information. We are particularly interested in benchmarking plural QF against vanilla QF and other mechanisms for funding public goods (like voting or direct contribution) under these circumstances. Another related area of work involves understanding to what extent a mechanism can be strategy-proof with respect to the expression of one's social groups. In other words, it could be fruitful to design mechanisms that naturally incentivize people to honestly report relevant social information.

C Groups can extract linearly growing subsidies out of Squared Cluster Match.

To show a scenario in which a group of agents can exploit Squared Cluster Match, we will keep things simple and use the same setup as in section 3.2. Suppose we have agents $\{1, 2, 3\}$ in groups $X = \{1, 2\}$ and $Y = \{1, 3\}$. Also, assume that $c_1 = c_2$. The funding amount under Squared Cluster Match is

$$\left(\sqrt{\frac{c_1}{4} + c_2} + \sqrt{\frac{c_1}{4} + c_3} \right)^2 \geq \frac{c_1}{2} + c_2 + \sqrt{\frac{c_1^2}{16} + \frac{c_1 c_2}{4}} \quad (27)$$

where the RHS was attained by expanding the power of 2 and removing terms that included c_3 . Letting c_X be the total amount contributed by all members of group X , we can re-write c_1 and c_2 as $c_X/2$. Then the RHS of (27) becomes

$$\frac{3c_X}{4} + \sqrt{\frac{5}{4} \cdot \frac{c_X^2}{16}} = \frac{3c_X}{4} + \sqrt{\frac{5}{4}} \frac{c_X}{4} = (1 + k)c_X \quad (28)$$

for a constant $k > 0$ (specifically $k = (\sqrt{5/4} - 1)/4$). So the total funding amount minus c_X is at least $k \cdot c_X$. In other words, as the members of X increase their contribution, the extra matching funds awarded to the project grow linearly. So Squared Cluster Match does not satisfy our definition of collusion resistance.

It is worth noting that the constant k above is small (around 0.023). More work is needed to see if that is just a quirk of this example or something more deeply true about Squared Cluster Match. With this group structure, at least, we cannot get k to be arbitrarily large: some calculus reveals the biggest k we can get is $k \approx 0.107$ by setting $c_1 \approx 0.2c_2$ rather than $c_1 = c_2$. In practice, it might be in the interest of mechanism designers reward groups who reveal themselves to the mechanism. So some mild linear increase in extra funds for groups, such as what we have identified here, might be a positive quality in some cases. By that same token, however, it is also important to note that with this group structure, when $c_1 \approx 1.5c_2$ or more k becomes negative, meaning that if c_3 small enough, the entire funding amount will be less than $c_1 + c_2$. In this case, it would not be rational for agents 1 and 2 to contribute anything at all (if possible, they would prefer to use another channel like PayPal to send the full amount $c_1 + c_2$ directly to the project). So more work is needed to explore where Squared Cluster Match incentivizes agents to report their group structures and where it discourages agents from participating in the mechanism.

D A rigorous definition of collusion resistance

For some QF mechanism M , let $M(c, G)$ be the amount of funding awarded to a project given contribution vector c and group structure G . Then we say that M is collusion-resistant if the following three sub-properties hold with respect to all agents i and all groups g :

1. *Diminishing returns for individuals*: $M(c, G) - c_i$ is $O(\sqrt{c_i})$. In other words, as agent i raises their contribution, the amount of subsidy capital given by M grows according to the root of their contribution.

2. *Diminishing returns for groups:* Suppose all members of G scale up their contributions by a multiplicative factor of β . Then $M(c, G) - c_g$ is $O(\sqrt{\beta})$. In other words, as the members of g increase their collective contribution, the subsidy capital grows according to the root of their collective contribution. To be clear, here c_g represents the combined contribution of group g *after* contributions have been scaled up.
3. *Diminishing returns from adding new members:* Suppose x new agents join group g , each contributing γ . These agents can be new to the system or can have previously been in other groups as well. Then $M(c, G) - c_g$ is both $O(\sqrt{x})$ and $O(\sqrt{\gamma})$. To be clear, here c_g represents the combined contribution of group g *after* the new members have been added.

Note that one could also expand this definition to simply require that the quantities above grow sub-linearly in the relevant variables: there is nothing special about using the square root here. However, this definition will make the proof in Appendix E more clean.

Although we imagine a certain uniformity over the behavior of agents (i.e. all agents in a group scale up their contribution by the same factor, or all new agents contribute the same amount), our definition of collusion resistance is still meaningful in scenarios with non-uniform behavior. For example, suppose that under a collusion-resistant mechanism M , the agents in some group g increased their contributions by different scalars. We could let β denote the largest scalar, and note that M 's collusion resistance gives guarantees on how funding would behave if all agents had increased their contribution by a factor of β – but that scenario is itself an upper bound on the actual scenario with disparate scalars. So, transitively, we get meaningful guarantees for settings with disparate scalars as well.

E Proof of Theorem 1: Connection-Oriented Cluster Match is collusion resistant

This section proves that Connection-Oriented Cluster Match satisfies the formal definition of collusion-resistance laid out above, in Appendix D.

Recall that given contribution vector c and group structure G , we can define the amount of funding awarded by Connection-Oriented Cluster Match as

$$COCM(c, G) := \sum_{g \in G} \sum_{i \in g} \frac{c_i}{|T_i|} + \sum_{g \in G} \sum_{h \neq g \in G} \sqrt{\sum_{i \in g} \frac{K(i, h)}{|T_i|}} \cdot \sqrt{\sum_{j \in h} \frac{K(j, g)}{|T_j|}} \quad (29)$$

where c_i is the contribution of agent i , T_i is the set of groups that agent i is in, and $K(i, h)$ is some attenuation of c_i depending on agent i 's connections to group h . In particular, if i is in h or if $i \notin h$ but i shares some group with some member of h , $K(i, h)$ is $\sqrt{c_i}$. Otherwise, $K(i, h)$ is c_i .

Now for the proof of the theorem.

Proof. First, note that

$$\sum_{g \in G} \sum_{i \in g} \frac{c_i}{|T_i|} = \sum_{i \in N} c_i \quad (30)$$

So for any agent i ,

$$COCM(c, G) - c_i = O(1) + \sum_{g \neq h \in G} \sqrt{\sum_{i \in g} \frac{K(i, h)}{|T_i|}} \cdot \sqrt{\sum_{j \in h} \frac{K(j, g)}{|T_j|}} \quad (31)$$

and similarly for any group g ,

$$COCM(c, G) - c_g = O(1) + \sum_{g \neq h \in G} \sqrt{\sum_{i \in g} \frac{K(i, h)}{|T_i|}} \cdot \sqrt{\sum_{j \in h} \frac{K(j, g)}{|T_j|}}. \quad (32)$$

Thus, in order to prove collusion resistance, it suffices to show that the sum of interaction terms grows at the appropriate rate. However, the number of groups is fixed, so the number of interaction terms is a constant. Therefore, showing that each individual interaction term grows at the appropriate rate suffices to prove the theorem. Now we will focus on each of the three sub-properties of collusion resistance independently.

Diminishing returns for individuals. Pick any agent $\hat{i}$ and any distinct $g, h \in G$. If agent $\hat{i}$ is in neither g nor h , $c_{\hat{i}}$ does not show up in the interaction term at all, so it is constant with respect to $c_{\hat{i}}$. If $\hat{i}$ is in just one group (w.l.o.g. suppose $\hat{i} \in g$, $\hat{i} \notin h$), then we have

$$\sqrt{\sum_{i \in g} \frac{K(i, h)}{|T_i|}} \cdot \sqrt{\sum_{j \in h} \frac{K(j, g)}{|T_j|}} = \sqrt{\frac{K(\hat{i}, h)}{|T_{\hat{i}}|}} + O(1) \cdot O(1) \quad (33)$$

Which is $O(\sqrt{c_{\hat{i}}})$ regardless of whether $K(\hat{i}, h)$ evaluates to $\sqrt{c_{\hat{i}}}$ or c_i . Lastly, if $\hat{i} \in g$ and $\hat{i} \in h$ then we know $K(\hat{i}, h) = \sqrt{c_{\hat{i}}}$, so the interaction term will be

$$\sqrt{\sum_{i \in g} \frac{K(i, h)}{|T_i|}} \cdot \sqrt{\sum_{j \in h} \frac{K(j, g)}{|T_j|}} = \sqrt{O(1)\sqrt{c_{\hat{i}}} + O(1)} \cdot \sqrt{O(1)\sqrt{c_{\hat{i}}} + O(1)} = \sqrt{O(1)c_{\hat{i}} + O(1)\sqrt{c_{\hat{i}}} + O(1)} \quad (34)$$

which is $O(\sqrt{c_{\hat{i}}})$.

Diminishing returns for groups. Pick any group $\hat{g}$. Now consider the interaction term between two groups h_1 and h_2 (possibly, one of these groups is $\hat{g}$ itself). If no member of $\hat{g}$ is in either group, then the interaction term is constant with respect to the members of $\hat{g}$ increasing their contribution. Next, suppose $\hat{g}$ has some overlap with one of the groups but not with the other: w.l.o.g. assume i.e. $|\hat{g} \cap h_1| \geq 1$ and $|\hat{g} \cap h_2| = 0$. Then we have

$$\sqrt{\sum_{i \in h_1} \frac{K(i, h_2)}{|T_i|}} \cdot \sqrt{\sum_{j \in h_2} \frac{K(j, h_1)}{|T_j|}} \leq \sqrt{c_g + \sum_{i \in h_1 \setminus \hat{g}} \frac{K(i, h_2)}{|T_i|}} \cdot \sqrt{\sum_{j \in h_2} \frac{K(j, h_1)}{|T_j|}} = \sqrt{c_{\hat{g}} + O(1)} + O(1). \quad (35)$$

Now if we fix the current group contribution level at c_g^0 and then scale up the contribution of each member of $\hat{g}$ by β (i.e. letting $c_{\hat{g}} = c_g^0 \cdot \beta$, then the far RHS of (35), which upper bounds the interaction term, will become $\sqrt{\beta c_g^0 + O(1)} + O(1)$, which is $O(\sqrt{\beta})$.

Lastly we will consider the case where $|\hat{g} \cap h_1| \geq 1$ and $|\hat{g} \cap h_2| \geq 1$. Here, we know that all members of $\hat{g}$ in this interaction term will have their contribution attenuated. Then examining the interaction term we have

$$\sqrt{\sum_{i \in h_1} \frac{K(i, h_2)}{|T_i|}} \cdot \sqrt{\sum_{j \in h_2} \frac{K(j, h_1)}{|T_j|}} \leq \sqrt{\sum_{i \in g} \sqrt{c_i} + \sum_{i \in h_1 \setminus \hat{g}} \frac{K(i, h_2)}{|T_i|}} \cdot \sqrt{\sum_{j \in g} \sqrt{c_j} + \sum_{j \in h_1 \setminus \hat{g}} \frac{K(j, h_1)}{|T_j|}} \quad (36)$$

$$\leq \left(\sqrt{\sum_{i \in \hat{g}} \sqrt{c_i}} + \sqrt{\sum_{i \in h_1 \setminus \hat{g}} \frac{K(i, h_2)}{|T_i|}} \right) \cdot \left(\sqrt{\sum_{j \in \hat{g}} \sqrt{c_j}} + \sqrt{\sum_{j \in h_2 \setminus \hat{g}} \frac{K(j, h_1)}{|T_j|}} \right) \quad (37)$$

$$= \sum_{i \in \hat{g}} \sqrt{c_i} + O(1) \sqrt{\sum_{i \in \hat{g}} \sqrt{c_i}} + O(1). \quad (38)$$

Now suppose that we fixed c_i as c_i^0 for every $i \in \hat{g}$ and then increased every member of $\hat{g}$'s contribution by a factor of β (i.e. letting $c_i = \beta c_i^0$ for $i \in \hat{g}$). Then (38), which upper bounds the interaction term, would become

$$\sum_{i \in \hat{g}} \sqrt{\beta c_i^0} + O(1) \sqrt{\sum_{i \in \hat{g}} \sqrt{\beta c_i^0}} + O(1) = \sqrt{\beta} \sum_{i \in \hat{g}} \sqrt{c_i^0} + O(1) \sqrt{\sqrt{\beta} \sum_{i \in \hat{g}} \sqrt{c_i^0}} + O(1) \quad (39)$$

which is $O(\sqrt{\beta})$.

Diminishing returns from adding new members. Suppose that x agents join some group $\hat{g}$ and donate γ each. Recall that these agents can be totally new to the system or can already be members of other groups, but here we only consider the effect of them joining group $\hat{g}$. For agents who were already in other groups, we imagine that they were donating γ before and after joining $\hat{g}$.

Because we only want to understand how funding changes as new agents join $\hat{g}$, and because agents who were already in other groups don't change their contribution amount, all interaction terms that don't directly involve $\hat{g}$ are constant with regard to x and γ . So we only need to examine interaction terms concerning $\hat{g}$ and some other group h .

If none of the x new members were in group h , then the interaction term evaluates to

$$\sqrt{\sum_{i \in \hat{g}} \frac{K(i, h)}{|T_i|}} \cdot \sqrt{\sum_{j \in h} \frac{K(j, \hat{g})}{|T_j|}} \leq \sqrt{x\gamma + O(1)} \cdot O(1) \quad (40)$$

where the constant term inside the square root represents the contributions of the members of $\hat{g}$ who were already there – this amount does not depend on x or γ .

Now suppose some of the x new members were in group h . This amount must be some constant y . Also note that since at least one agent is in both $\hat{g}$ and h , $K(i, h)$ and $K(j, \hat{g})$ will evaluate to $\sqrt{c_i}$ and $\sqrt{c_j}$ for all $i \in \hat{g}$ and $j \in h$. Now, the interaction term will be

$$\sqrt{\sum_{i \in \hat{g}} \frac{K(i, h)}{|T_i|}} \cdot \sqrt{\sum_{j \in h} \frac{K(j, \hat{g})}{|T_j|}} \leq \sqrt{x\sqrt{\gamma} + O(1)} \cdot \sqrt{O(1)\sqrt{\gamma} + O(1)} \quad (41)$$

$$= \sqrt{O(1)x\gamma + O(1)\sqrt{\gamma} + O(1)x\sqrt{\gamma} + O(1)} \quad (42)$$

which is $O(\sqrt{\gamma})$ and $O(\sqrt{x})$.

□

F Results on Offset Match

Recall that $s_{i,j} = |T_i \cap T_j|/|T_i|$ and we're looking for coefficients α_i satisfying $\sum_{j \in N} \alpha_j \cdot s_{j,i} = 1$ for all $i \in N$. Once we have those values, the funding amount given by Offset Match will be $(\sum_{i \in N} \sqrt{\alpha_i c_i})^2$.

F.1 Proof of Theorem 2: Offset Match feasibility is characterized by linear dependence among group profile vectors

To formalize a little more, let S be a square matrix where the i th column of S is $(s_{1,i}, s_{2,i}, \dots, s_{n,i})$ and $S_{i,j} = s_{i,j}$. Then finding the α_i coefficients is equivalent to finding a vector α such that $S^T \alpha = \mathbf{1}$ (where $\mathbf{1}$ is the vector of n 1's in a row). Like in any system of equations, there will be exactly one solution if and only if S has full rank.

Let agent i 's *group profile vector* be the vector $(1_{i \in g_j})_{g_j \in G}$, i.e. the vector has a 1 in the j th position if agent i is in group j , and a 0 otherwise.

Now we can rephrase Theorem 2 as follows: *$S^T \alpha = \mathbf{1}$ has a solution (and Offset Match is usable) if and only if there is linear independence dependence among the set of all agents' group profile vectors.*

Proof. Let A be a matrix with m rows and n columns, where $a_{i,j} = 1_{i \in \sigma_j}$ (So the i th row of A is i 's group profile vector). Let $B = AA^T$. Then B is an $n \times n$ matrix with $b_{i,j} = |T_i \cap T_j|$. Note that one can turn B into S by executing the following elementary row operations: set every entry in the i th row equal to itself divided by $|T_i|$. For any matrix X over the reals, $\text{rank}(X) = \text{rank}(XX^T)$, and elementary row operations preserve rank. So $\text{rank}(A) = \text{rank}(S)$.

Linear independence among group profile vectors is equivalent to linear independence among the rows of A . If this occurs, then A (and therefore S) will rank n , meaning the system $S^T \alpha = \mathbf{1}$ has exactly one solution. Otherwise, A and S will have rank less than n , meaning that the system does not have exactly one solution. $\square$

Corollary 1. *If $|G| < |N|$, Offset Match is guaranteed to be infeasible.*

Proof. In this case, A will have less than n rows. For any matrix A , the rank of A is at most $\min\{\# \text{ rows in } A, \# \text{ columns in } A\}$. So $n > \text{rank}(A) = \text{rank}(S)$. $\square$

Corollary 2. *If one puts each agent in a dedicated singleton group including only themselves (in addition to other groups they may be in), then Offset Match is guaranteed to be feasible.*

Proof. In this case, every agent's group profile vector will have a 1 in a unique spot (the spot corresponding to their singleton group), in which all other vectors only have zeroes. So linear dependence is impossible. $\square$

It remains unclear how putting every agent in a singleton group would affect α values.

F.2 Scenarios exist where almost all agents have $\alpha_i = 0$

Recall that if Offset Match is feasible for a given group structure, the vector α describes to which extent each agent's contribution is scaled down when calculating funding. One can construct a scenario where all but one element of α is 0. Although the group structure constructed to show this is unrealistic, it signals the need for more work on how α reacts to group structure.

Lemma 1. *For all $n \geq 1$, a group structure G on agents $N = [n]$ exists such that Offset Match is feasible but $\alpha_i = 0$ for all $i > 1$.*

Proof. Let $G = \{g_1, \dots, g_n\}$ and let $T_i = \{\sigma_j | j \leq i\}$ for all $i \in N$. So agent 1 is just in g_1 , agent 2 is in g_1 and g_2 , and so forth, and agent n is in all n groups.

Under those conditions, one can calculate entry $s_{i,j}$ of the matrix S as

$$s_{i,j} = \begin{cases} 1 & \text{if } i \leq j \\ \frac{j}{i} & \text{if } i > j \end{cases} \quad (43)$$

In other words, the i th row of S has entries that start at $s_{i,1} = 1/i$ and gradually increase up to the i th entry, which will be 1, and all subsequent entries will be 1s. So if one fixes a column j and looks at the next column to the right, every column variable will either stay the same or get larger. In contrast, if one looks at the column to the left of j , every row variable will either stay the same or decrease.

We should now be clear about what this means for the structure of S^T since that is the matrix used in the system of equations. If one fixes a row j in S^T and looks at the next row down, every row variable will either stay the same or get larger. In contrast, if one looks at the row above j , every column variable will either stay the same or decrease. Moreover, note that the lower-right triangle of S^T is all 1s.

This structure for S^T means that the only vector α satisfying $S^T \alpha = \mathbf{1}$ has a 1 in the first entry and zeroes in the last $n - 1$ entries. Suppose there is some other vector $\alpha' \neq \alpha$ and a row vector j (taken from the j th row in S) such that $j \cdot \alpha' = 1$. Then due to the monotonic way in which the values in columns change when you move up and down in the matrix, that same vector will yield $j' \cdot \alpha < 1$ for some rows $j' < j$ (i.e., some row above j in the matrix) and $j' \cdot \alpha > 1$ for some row $j' > j$.

Indeed, this group structure yields a system of equations with a solution, but that solution vector has $\alpha_i = 0$ for all $i > 1$. $\square$

This construction suggests that something fishy might occur whenever we have i, j with $T_i \subset T_j$. More work is needed to confirm if this is the case.